Jonny Evans

Machine Learning Engineer

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Experienced Machine Learning Engineer with a background in AI, data science and mathematics. Expert in turning business problems into impactful AI based systems.

PROFESSIONAL EXPERIENCE

Deliveroo, London (UK) September 2024 - Present

Senior Machine Learning Engineer

- Leading the development of our contextual bandit model responsible for determining ad-load throughout the app. Included development of neuralUCB predictors within the bandit, development of new features, and launching AB tests.
- Implemented many SQL query optimisations, saving over £150k in annual compute costs.
- Built and maintained complex data pipelines supporting our ML models, processing multi-billion-row tables.
- Mentoring and buddying junior colleagues.
- Tech included Python, Snowflake, DBT, AWS, Terraform, Looker, DataDog, Go.

Carbon Re, London (UK) May 2023 - June 2024

Machine Learning Engineer

- Led the ML team in rolling out AI capabilities to new clients.
- Designed and implemented a state-machine model to infer the overall status of a cement plant, used by various downstream ML applications.
- Designed and implemented an extensive outlier cleaning system used by all our ML models.
- Taken multiple machine learning models from business idea to production.
- Tech included Python, Docker, AWS, Terraform and Streamlit.

Shift Lab, London (UK) June 2021 - April 2023

Machine Learning Engineer

- First employee at an Applied AI lab led by the ex head of DeepMind Health. Helped grow the team from 3 to 30+ by interviewing, mentoring, and leading project teams.
- Project lead for a quality assurance (QA) platform based on fine-tuned SOTA image recognition algorithms. Involved building and testing a web application that allowed users to upload and manage images, receive image detections and QA recommendations, and re-label images (closing the ML feedback loop). Trialled by QA engineers in the Arrival electric vehicle factory, who chose to use the system in full production.
- Fine-tuned and implemented LLMs and image generation models for a web application that let users build brands from scratch. Responsible for the entire process of literature research to deployed models for various AI features and systems within the application. This included research, experimentation, user testing, managing stakeholders across disciplines, marketing domain knowledge, engineering and MLOps.
- Designed, developed, and deployed lease price forecasting algorithms for a shipping container buy/lease/sell trading platform.
- Tech included Python, Docker, AWS, MongoDB, and Streamlit.

Deliveroo, London (UK) September 2019 - June 2021

Machine Learning Engineer II

- Designed, tested, and implemented new delivery fee optimisation algorithms, with AB test results showed a 1.5% increase in profit per order (worth ~£1.5m annually) with insignificant order volume impact.
- Led the rollout of the above algorithms to 9 of Deliveroo's key markets, and provided mentorship and training to 3 colleagues on how to use the algorithm and simulate its effects using shadow pricing and price elasticity models.
- Implemented deep learning interpretability methods and used them to inform changes to our restaurant recommendation algorithms.
- Developed regression models on very tight timelines to inform key business decisions on changes to delivery distance limits.
- Built and maintained robust data pipelines that serviced our ML models.
- Tech included Python, SQL, Docker, AWS, Snowflake, Looker, Jenkins, DBT, and CircleCI.

University of Southampton (UK) July 2016 – August 2019

Ph.D. Machine Learning – Funded by Siemens Mobility Limited and EPSRC.

- Developed machine learning algorithms for road traffic forecasting and incident detection using novel contextual features, resulting in improvements over the state-of-the-art in the forecasting algorithm of 4.4% mean squared error, and in the incident detection algorithm of 39% detection rate and 37% false alert rate.

- Implemented the algorithms within Siemens' software products, which involved pitching, writing successful funding bids, cross-discipline collaboration, and pair-programming with a Lead Software Architect. This led to the algorithms being implemented within TfL and various city council's versions of Siemens' products.
- Conducted a month-long trial of the algorithms in Bristol Council's Traffic Management Centre, wherein the algorithms detected 11 otherwise unknown incidents.
- Designed and built a web application for the above trail that displayed alerts to operators and received feedback in real time.
- Advised two Siemens interns and an MSc student in research projects to improve the above traffic forecasting algorithm, funded by the Department For Transport. Siemens continued this work long after the PhD was completed.
- Interviewed Traffic Management Centre managers and operators on the state of incident detection in the U.K.
- Ph.D. project featured in University's Re:action magazine (page 13).
- 5 publications in leading transportation journals and global conferences, detailed here.
- Tech included Python, Scikit-learn, SQL, Docker, AWS, Flask, JavaScript, HTML, CSS, APIs, and web sockets.

Atkins, Birmingham (UK) September 2015 – June 2016
Graduate ITS Consultant
- Built a Visual Basic for Applications (VBA) based forecasting model of household recycling for Hampshire County Council policy advisors.
- Analysed safety, traffic, and geometry data to find the optimal locations to implement ramp meters in North Carolina, U.S.A. Co-authored the final report to the North Carolina Department of Transportation, which recommended the 30 most suitable ramp meter locations for implementation.
- Acted as Head of Stakeholder Engagement for a feasibility study regarding the deployment of temporary electric vehicle rechargers across the U.K. for Highways England. Co-authored the final report to Highways England, which was subsequently published in the 11th ITS European Congress.
- Managed a team of 3 graduates and apprentices in a project which assessed the use of Smart Motorways in the U.K.

University of Birmingham (UK) September 2012 - September 2015
BSc Mathematics (First)
- Final year Research Project: 'Google's PageRank algorithm', won the Blackburn prize for the highest mark in graduating year.
- Social Secretary of the University table tennis club.

Cirrus logistics, Basingstoke (UK) June 2014 - August 2014
Intern
- Tested and developed our software product that optimised the scheduling of shipping berths.
- Created new Visual Basic for Applications (VBA) based PowerPoint templates for client training materials.

KEY SKILLS

Programming
Python, AWS, EC2, S3, Sagemaker, CloudWatch, Timestream, Docker, Kubernetes, Terraform, SQL, MongoDB, C#, Snowflake, Git, GitHub, Spark, CircleCI, DataDog, GitHub Actions, DBT, Flask, Tensorflow, Pytorch, GPT, Stable Diffusion, Llama, Pandas, NumPy, Seaborn, Matplotlib, Plotly, Looker.

Theory
Deep learning, tree-based machine learning, NLP, Large Language Models (LLMs), image generation, optimisation, AB testing, statistics, regression.

Communication
Project and technical leadership, mentor, interviewer, buddy, cross-discipline collaborator, presenter, technical and academic writer.

PERSONAL

- Built a portfolio of small projects on GitHub. Includes AutoPro, a chrome extension that improves the AutoTrader search experience. And SymbolFix, a Visual Basic Excel add-in that automates the formatting of symbols for Mechanical Engineers, and is still being used regularly today by ex-colleagues at Atkins.
- Keen table tennis player since the age of nine. Once ranked 2nd in England (under 13s). Professionally coached players of various ages and abilities.